Uncertainty-Guided Multi-Scale Boundary Refinement Network for Small Camouflaged Object Detection

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GitHub Repository: https://github.com/Shafi540/COD_SmallObject.git

I. LITERATURE SURVEY

A. Introduction to Literature Survey

Camouflaged Object Detection (COD) is a computer vision task concerned with segmenting objects that are deliberately or naturally blended into their surrounding environment, making them far harder to detect than objects in standard object detection or salient object detection tasks. Since the release of large-scale benchmark datasets such as COD10K and CAMO, deep learning-based approaches have made rapid progress on this problem, but persistent challenges remain around blurred object boundaries, small or multiple camouflaged objects, and overconfident predictions in ambiguous regions. A literature review is necessary at this stage of the project to understand which architectural strategies (boundary guidance, multi-scale fusion, uncertainty modeling) have already been explored, to identify which combinations remain underexplored — particularly for small camouflaged objects — and to ensure the proposed uncertainty-guided, multi-scale boundary refinement network is positioned as a genuine, well-motivated contribution rather than a duplication of existing work.

B. Paper-wise Analysis

1) 1. Yan, Sun, Han, and Wang (2024): Objective: To unify localization, recognition, and precise boundary segmentation of camouflaged objects within a single sequential pipeline, rather than treating these as separate stages.
Dataset/Environment: Standard COD benchmark datasets (COD10K, CAMO, NC4K).
Methodology/Model: Yan et al. (2023) developed a matching–recognition–refinement network, where candidate camouflaged regions are matched, the object class is recognized, and the segmentation boundary is refined in successive stages, published in IEEE Transactions on Neural Networks and Learning Systems.
Key Findings: The authors reported competitive segmentation accuracy relative to existing COD baselines across standard evaluation metrics.
Limitation: A potential limitation is error propagation across the sequential matching, recognition, and refinement stages, although this is an architectural inference rather than an author-reported limitation.
2) 2. Xia, Cao, Gao, Ge, Li, Fang, Zhang, and Liang (2024a): Objective: To address coarse boundary identification in complex scenes and the ineffective integration of multisource features in existing COD networks, with particular attention to small and multiple camouflaged objects.
Dataset/Environment: Standard COD benchmark datasets.
Methodology/Model: Xia et al. (2024a) proposed BERNet, consisting of an asymmetric edge module with a multigroup dilated convolution block for rich boundary cues, a residual mixed pooling enhanced module, and a multivariate information interaction refiner module, published in Machine Vision and Applications.
Key Findings: The authors reported improved boundary accuracy and detection performance relative to prior boundary-based COD methods, with particular gains on small and multiple-object scenes.
Limitation: A potential limitation is increased architectural complexity from the multi-group dilated-convolution branches and multiple refinement modules; this is an architectural inference rather than an author-reported limitation.
3) 3. Shi, Ren, Chen, Zhao, Lin, and Zhao (2023): Objective: To address the inefficient use of global context information

and boundary cues in prior COD architectures. Dataset/Environment: Four benchmark COD datasets. Methodology/Model: Shi et al. (2023) proposed a context-aware and boundary refinement (CABR) network consisting of a global context information enhanced module, an attention-inducing neighbor fusion module for multi-scale and small-object detection, and a boundary refinement module, published in Applied Intelligence.

Key Findings: The authors reported performance competitive with existing state-of-the-art COD methods across all four datasets.

Limitation: A potential limitation is that the multi-module context, fusion, and boundary-refinement design increases architectural complexity compared with simpler single-branch COD networks.

4) 4. Ge, Ren, Zhang, He, Bi, and Zhang (2025): Objective: To reduce the loss of fine object detail that occurs when camouflaged object features are processed only at a single resolution scale.

Dataset/Environment: Standard COD benchmark datasets. Methodology/Model: Ge et al. (2024a) proposed a feature-aware and iterative refinement network that progressively refines the predicted segmentation mask across multiple iterations, using feature-awareness modules to emphasize informative regions at each step, published in The Visual Computer.

Key Findings: The authors reported progressive improvement in mask accuracy across successive refinement iterations compared to single-pass COD baselines.

Limitation: A potential limitation is increased inference latency from the iterative refinement process; this is an architectural inference rather than an author-reported limitation.

5) 5. Ge, Zhong, Ren, He, Bi, and Zhang (2024b): Objective: To make explicit use of object location within a scene as a detection cue, which prior fusion-based COD networks underuse.

Dataset/Environment: Standard COD benchmark datasets. Methodology/Model: Ge et al. (2024b) developed a location-awareness and feature fusion network that models spatial location priors alongside standard multi-level feature fusion, published in Image and Vision Computing.

Key Findings: The authors reported measurable improvement over fusion-only baselines that do not incorporate explicit location priors.

Limitation: The method can still struggle in complex environments where objects and backgrounds have highly similar textures, making precise edges and structures difficult to distinguish.

6) 6. Ren, Zhang, Kang, Zhong, He, Ge, and Bi (2025): Objective: To address the underuse of combined semantic and spatial cues across multi-level features in existing COD networks.

Dataset/Environment: Standard COD benchmark datasets. Methodology/Model: Ren et al. (2025) proposed the Semantic-spatial guided Context Propagation Network (SCP-Net). It uses a twin positioning module for semantic cues, a spatial awareness module for shallow spatial cues, and a context propagation module (CPM) to assign semantic and spatial cues to multi-level features, published in Applied Intelligence.

Key Findings: The authors reported improved performance over baseline fusion networks that do not explicitly separate semantic and spatial guidance.

Limitation: A potential limitation is additional architectural complexity from the separate semantic, spatial, and context-propagation modules; this is an architectural inference rather than an author-reported limitation.

7) 7. Zhang, Zhang, Hamidouche, and Déforges (2023): Objective: To address model bias and data bias in COD, where center-biased training data causes poor generalization and boundary labels remain inherently ambiguous.

Dataset/Environment: COD10K, CAMO, CHAMELEON, and NC4K.

Methodology/Model: Zhang et al. (2023) proposed PUENet, built around a Bayesian conditional variational autoencoder (BCVAE) combined with a predictive uncertainty approximation (PUA) module, published in IEEE Transactions on Image Processing.

Key Findings: The authors reported performance competitive with or exceeding prior state-of-the-art COD models across all four benchmark datasets.

Limitation: Although PUENet uses a variational uncertainty framework, its predictive uncertainty approximation (PUA) module is specifically designed to avoid the expensive sampling process at test time.

8) 8. Zhai, Li, Yang, Jiao, Luo, Cheng, and Liu (2023): Objective: To address the limitation of prior COD methods that treat camouflaged regions independently without modeling relationships between different image regions.

Dataset/Environment: CHAMELEON, CAMO, COD10K, and NC4K.

Methodology/Model: Zhai et al. (2023) developed a mutual graph learning (MGL) framework separating object region reasoning and edge reasoning, connected through a mutual graph module, published in IEEE Transactions on Image Processing.

Key Findings: The authors reported improved performance over prior graph-free COD baselines on standard evaluation metrics.

Limitation: A potential limitation is the additional architectural and optimization complexity introduced by mutual graph reasoning between object and edge representations.

9) 9. Liu, Qiu, Yu, Li, and Li (2023): Objective: To address coarse boundary identification and insufficient multi-level feature fusion in existing COD methods.

Dataset/Environment: CAMO, CHAMELEON, and COD10K.

Methodology/Model: Liu et al. (2023) proposed MFNet, consisting of an edge guidance module, a multi-level feature integration module, and a context aggregation refinement module, published in Sensors (MDPI).

Key Findings: The authors reported that MFNet outper-formed prior state-of-the-art models across all four evaluation metrics (S-measure, E-measure, weighted F-measure, MAE).
Limitation: A potential limitation is increased architectural complexity from combining edge guidance, multi-level feature integration, and context aggregation refinement modules.
10) 10. He, Li, Zhang, Tang, Zhang, Guo, and Li (2023): Objective: To address camouflaged objects whose visual features are difficult to separate from background clutter using standard feature extraction alone.
Dataset/Environment: COD10K, CAMO, CHAMELEON, and NC4K.
Methodology/Model: He et al. (2023) proposed a feature decomposition and edge reconstruction framework, separating features into distinguishable and less-distinguishable components before applying dedicated edge reconstruction, published at CVPR.
Key Findings: The authors reported state-of-the-art results at the time of publication across standard COD metrics.
Limitation: Despite its feature-decomposition and edge-reconstruction components, the authors report cheaper computational and memory costs than the compared state-of-the-art methods.
11) 11. Ye, Li, Huo, et al. (2025): Objective: To improve COD in challenging situations such as occluded targets and scenes with multiple targets through boundary-guided multi-scale refinement.
Dataset/Environment: Three benchmark COD datasets.
Methodology/Model: Ye et al. (2025) proposed BGMR-Net, incorporating a Global Information Extraction (GIE) module and a Boundary-Aware (BA) module that integrates features across all scales, published in The Visual Computer.
Key Findings: The authors reported strong performance relative to prior boundary-guided COD networks.
Limitation: The paper identifies challenging cases such as occluded targets and scenes with multiple targets; increased GPU/memory demand is not established as an author-reported limitation.

C. Comparative Summary Table

Author & Year Technique Dataset Performance Metrics Limitation
Potential error propagation across stages (inferred)
Yan et al. (2024) Matching–recognition– refinement network COD10K, CAMO, NC4K Standard COD metrics

Xia et al. (2024a)	Asymmetric edge + dilated conv + refiner (BERNet)	Standard COD benchmarks	Standard COD metrics	Potential multi-module/parallel-branch complexity (inferred)
Shi et al. (2023)	Global context + boundary refinement (CABR)	4 COD benchmarks	Standard COD metrics	Multi-module architectural complexity
Ge et al. (2025)	Feature-aware iterative refinement	Standard COD benchmarks	Standard COD metrics	Potential iterative-refinement latency (inferred)

Fig. 1. Literature survey comparative summary (part 1)

Within the reviewed literature, this indicates a logical gap: the reviewed methods do not combine uncertainty estimation, multi-scale feature fusion, and uncertainty-guided boundary refinement into a single framework evaluated specifically for small camouflaged objects. The boundary/multi-scale methods reviewed here do not use predictive uncertainty to guide boundary refinement, while PUENet (Zhang et al., 2023) estimates predictive uncertainty but does not use it as an explicit control signal for a dedicated multi-scale boundary-refinement process. This gap motivates the proposed uncertainty-guided, multi-scale boundary refinement network for small camouflaged object detection.

Author & Year	Technique	Dataset	Performance Metrics	Limitation
Ge et al. (2024b)	Location-awareness + feature fusion	CAMO, COD10K, NC4K	Standard COD metrics	Difficulty with highly similar textures/complex scenes
Ren et al. (2025)	Semantic-spatial context propagation (SCPNet; CPM module)	3 COD benchmarks	Standard COD metrics	Potential multi-module complexity (inferred)
Zhang et al. (2023)	Bayesian CVAE + uncertainty approximation (PUENet)	COD10K, CAMO, CHAMELEON, NC4K	S-measure, E-measure, weighted F-measure, MAE	PUA avoids expensive test-time sampling
Zhai et al. (2023)	Mutual graph learning (MGL)	CHAMELEON, CAMO, COD10K, NC4K	S-measure, E-measure, MAE	Graph-reasoning complexity
Liu et al. (2023)	Edge guidance + multi-level fusion (MFNet)	CAMO, CHAMELEON, COD10K	Sα, Eϕ, Fβw, MAE	Potential architectural complexity
He et al. (2023)	Feature decomposition + edge reconstruction	COD10K, CAMO, CHAMELEON, NC4K	Standard COD metrics	Authors report cheaper compute/memory costs
Ye et al. (2025)	Global info extraction + boundary-aware fusion (BGMR-Net)	3 COD benchmarks	Standard COD metrics	Occluded and multiple-target scenes remain challenging

D. Research Gap

Across the reviewed papers, three consistent patterns emerge. Boundary refinement and multi-scale feature fusion are dominant strategies, appearing in Shi et al. (2023), Xia et al. (2024a), Liu et al. (2023), Ren et al. (2025), and Ye et al. (2025), but these works do not jointly optimize boundary sharpening and scale-awareness alongside an explicit measure of prediction confidence. Small and multiple camouflaged objects are explicitly considered in works such as Shi et al. (2023) and Xia et al. (2024a), yet small-object performance is generally not treated as a dedicated primary evaluation objective. Uncertainty modeling appears in Zhang et al. (2023), where PUENet models model bias and data bias, including ambiguity around object boundaries; however, the estimated uncertainty is not used explicitly to determine where a boundary-refinement module should concentrate its refinement effort

Fig. 2. Literature survey comparative summary (part 2)

II. METHODOLOGY

The output of a camouflaged object detection model can be strongly influenced by the diversity and the quality of the training and evaluation data that have been employed. In our project we present open source camouflaged detection data sets and investigate how good a model can learn to recognize something that is camouflaged. All data sets are provided with RGB images and ground-truth masks (which show the ground-truth segmentation at pixel-level).
• COD10K: COD10K is a large-scale camouflaged detection dataset containing 10,000 images spanning 78 object categories. It includes camouflaged and non-camouflaged images, along
with detailed interpretations such as object masks and bounding objects. Out of those 10,000 images, 5,066 images consist of camouflaged objects. The commonly used camouflaged subset consists of 3,040 training images and 2,026 testing images. These two qualities of this dataset, i.e., range and annotation diversity makes it an appropriate primary dataset for evolving and verifying camouflaged object detection models.
- CAMO: The 1,250 images of natural and synthetic camouflaged objects of the CAMO dataset.
The normal division contains 1,000 training images and 250 testing images. It has ground truth of object-level masks on

the dataset and includes difficult scenes where the foreground objects are similar to the background objects, which is helpful to test the effectiveness of extracting camouflaged objects with complex backgrounds.

- NC4K: NC4K is a large-scale challenging camouflaged object detection benchmark. The NC4K dataset comprises 4,121 images with ground-truth masks. It offers additional benchmark.

Data for model accuracy analysis of classifying many objects and a dozen scenes.

- Chameleon: Chameleon is a small camouflaged object detection dataset composed of 76 images and manually annotated ground-truth masks. It is potentially able to offer an additional Baiqu benchmark for testing the model ability of in challenging camouflaged situations. The above data sets is incorporated various instances of image and ground-truth segmentation mask for camouflaged object detection (COD) for the training and testing respectively, to meet the proposed experimental setting.

Model. The model is validated through estimation by the predicted segmentation mask against their ground truth segmentation mask.

A. Preprocessing Details

Preprocessing is a key step to process the image data for the camouflaged object detection model. The sample code defines the preprocessing pipeline as:

- The algorithm is implemented and the algorithm searches the dataset folder for Image files of supported extension-JPG, Jpeg, PNG etc. Supported All the above mentioned file extensions are supported by OpenCV library. Respective

Correcting ground-truth masks are retrieved using filename patterns and the normalized file names. It is important that every image be paired with the correct object mask so that the model can learn from correct input/output pairs.

- Image Resizing: The entire image, as well as the masks, are resized to a constant fixed resolution of 352 × 352 pixels. The resizing is performed using a bilinear-style resampling of the RGB image and the nearest-neighbor resampling of.

The difference above enables us to keep the foreground and background masks as integers.

- Data Augmentation: Data augmentation is performed using the following methods random horizontal flip (p=0.5), vertical flip

If rotation is used then the angle is sampled randomly between. -10and +10. In addition, 0.3 probability of random brightness and exposure for RGB image are brightened with brightening coefficient 0.85-1.15. Spatial distortion and brightening change operation is to change the operation on the image and the mask, and the change operation is to change on the image alone. These operations are to improve the generalization ability of the model.

- In this step, the RGB images are converted to Pytorch tensors and modified with Image Net channel means (0.485, 0.456, 0.406).

TABLE I
SUMMARY OF DATASETS REFERENCED IN THE PROJECT

Dataset	No. of images	Purpose
COD10K	10,000	Primary dataset referenced for training and testing
CAMO	1,250	Additional benchmark
NC4K	4,121	Additional evaluation benchmark
CHAMELEON	76	Small-scale evaluation benchmark

Fig. 3. Compilation of all data sets referred to in the project.

Standard deviations (0.229, 0.224, 0.225). This pre-processing supports the usage of a Image Net-pretrained ResNet34 backbone. We translate the ground truth masks to 1-channel tensors, thresholds at 0.5 to provide binary labels (target pixels:1, background pixels:0).

- The reference coding was for batch size 8. The prepared images and masks can be batched for training and validating the model using.

The data loaders. This code attempts to make use of the training, validation and test directories that are present, if they are present. If the code cannot find an appropriate official split for COD10K, then it creates a deterministic fallback split based on a fixed random seed. The relevant split ratios are different on each branch of the implementation. These three pre-processing steps should be simply considered to be information for the actual data-pipeline that takes place afterwards. It is good practice to confirm in the final experimental report that the datasets loaded successfully, along with the relevant data preprocessing and data-splitting steps, before reporting training or review results.

B. Feature Engineering and Selection

Column-fusion based intrusion-detection systems usually adopt feature selection for columns, which is irrelevant to COD since the input is an RGB image with pixel-wise binary mask. We build visual features instead of selecting columns for this case. We combine multi-scale features, boundary cues, object-size cues and pixel-wise uncertainty learned from CNN to discriminate small camouflaged objects and their backgrounds. 1) Multi-scale visual feature extraction: The encoder produces multi-scale feature maps of each normalized image. The lowest features encode texture and local contrast and detail of edges; intermediate features encode the arrangement of shape and parts; later features encode semantics. The different resolution scale features of the encoder are matched to the same number of channels and summed after they are interpolated to the same resolution, preserving the evidence for small objects which would be lost after successive down samples. 2) Boundary-aware feature engineering: Ground-truth boundaries

are extracted from binarized masks and dilate/erode to generate boundaries:

B = dilate(M) -erode(M), (1)

where M is the binary mask. Boundary targets supervise the second auxiliary edge prediction head. The shallow high-res feature stream is fused with the decoder features in the decoder head to restore the tiny structures as well as to localize weak object-background transitions. The morphological kernel size is to be specified and kept constant during experiments.
3) Object-size-aware encoding: R for each training mask (0.15 and 0.4 respectively). Images where r ≤0.10 are regarded as part of the small-object subset used in the analyses of the current project. Object-size ratios are supplied for stratified reporting, and (optionally) for sampling/loss weighting, they apply solely to the training set, but not to the test distribution.
4) Uncertainty feature construction: Segmentation head also predicts p ∈[0, 1] class probabilities at pixel level. Predictive entropy
stores the normalized value of predictive uncertainty (divided by log 2). It is highest at p = 0.5 and lowest at the over-confident, correctly predicted objects. By concatenating this uncertainty map to the decoder features and the coarse segmentation prediction we obtain an input-dependent prediction of a refinement factor for each potentially confused edge location. The amount of error correction should depend on the amount of the uncertainty, but the single sample entropy does not include this information as it does not incorporate knowledge of the underlying distribution of predictions (there must be an ensemble of many stochastic predictions to estimate this).
5) Feature fusion and selection strategy: Rather than using exactly an Information Gain or something similar with a predefined list of top-N pixels, the channel and spatial salience is learned via the trainable attention or gating module. For each scale, the channel gate (scores assigned for each of the feature channels) and the spatial gate (foreground probable, ambiguous boundary inside image) are presented. The feature maps after the gating are merged before it is input to the decoder. 6) Ablation study for validation: Features reported are contributions ablation, baseline encoder decoder, +multi-scale fusion, +boundary supervision, +uncertainty refinement, +full network. Same splits, resolution, optimizer, random seeds, and code evaluation. Report full test set and small-object subset from fig2 with the Dice or F1, the IoU, the MAE, and the boundary F1. Trade-offs the computational cost, parameters, inference time, memory, and ablation should have one.

C. Selection of the Algorithm and Justification

1) Chosen Method: We choose our deep encoder decoder segmentation network with multi-scale fusion, boundary supervision and uncertainty-guided refinement as the baseline network. Our network contains four modules, the encoder encodes semantic information and details, the decoder gradually up-samples the prediction, the auxiliary boundary branch emphasizes the object edge, and the uncertainty-guided module refines the uncertain prediction. The entire foreground probability map can be obtained from our network, and in case
of binary segmentation task, this probability map can also be binarized with a threshold. 2) Justification against other options: Traditional thresholding and hand-crafted features. Color thresholds, texture descriptors and classical edge detectors are influenced by

$$U = \frac{-p\log(p+\epsilon) - (1-p)\log(1-p+\epsilon)}{\log 2} \qquad (2)$$

Fig. 4. Predictive entropy

illumination and low contrast. They don't manage to reach the context difference of the object blended with the background. Typical CNN encoder-decoder / U-Net. A standard encoderdecoder: a reproductible baseline but the repeated down sampling does not keep the small-object details. Skip connection can recover details; explicit multi-scale aggregation and boundary losses are suggested to improve the details recovery. Transformer-based segmentation. Sure, global attention can learn long-range relations, but requires a lot of training data and GPU time. CNN or hybrid backbone is then a good compromise between the granularity of localization, pretrained representation and GPU resources. In this way select your backbone carefully, and don't use the latter blindly. Refinement with multi-scale and boundary-awareness. Multi-scale fusion reduces the scale of objects; explicit boundary targets address the fuzzy borders of segmentation; and the uncertainty-guided refinement increases the capacity at wrong pixels. Their combination speaks for our project problems: small objects, camouflaged objects, and the partial mask fuzzy border. 3) Training objective: A candidate composite objective might be:
Ltotal = λsegLseg + λboundaryLboundary + λrefLref, (3)
where Lseg is a linearly-merged binary cross-entropy with soft Dice loss on the main mask, Lboundary is binary cross-entropy
Figure 1 shows the end-to-end data flow of the proposed system, from the raw benchmark data to the final performance metrics. The workflow begins with the COD10K-v3 and
or soft Dice loss on the boundary target, and Lref supervises the refiner mask to the original ground-truth mask. We should report the "a priori" values of each λ and deep-supervision weight along with the optimizer and learning-rate schedule. As always, the weights are tuned on the validation set, not on the hold-out test set. 4) Experimental comparison and selection criteria: Train and test our network against an encoder-decoder baseline on the same train/validation/test splits (and input size). Select checkpoints based on the validation dice or a pre-declared combined validation score. Small-object dice/f1 is everything – other dice, iou, mae, boundary f1 can help contextualize. Also report the mean/stddev of each metric on multiple random seeds where relevant. Count parameters, FLOPS, or latency to determine whether additional complexity for refinement is worth it. 5) Reproducibility and status of implementation: This section outlines the general
approach suggested, not the fact that all components have been implemented and verified experimentally. Final submission-replace architecture decisions with the implemented back-

bone, feature-map strides and channels, fusion method, boundary kernel, loss weights, input resolution and augmentation pipeline, random seeds and software version used for the experiments performed. Present baseline results apart from what is proposed to be improved.

D. System Workflow / Block Diagram

CAMO-V.1.0 datasets, from which only camouflaged images are retained (the NonCAM images of COD10K are excluded). Each image is then paired with its ground-truth mask by

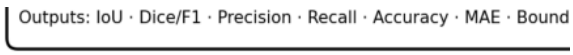


Fig. 5. Training objective

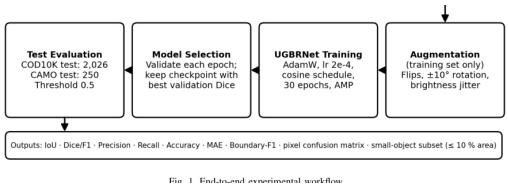


Fig. 6. End-to-end experimental workflow

file name, which acts as the data-cleaning step because it guarantees valid input-target pairs. In the preprocessing stage, images and masks are resized to 352 × 352 pixels, masks are binarized, and images are normalized with ImageNet statistics. The pooled data is then split, with 10% of each dataset's training images held out for validation (3,636 training and 404 validation images). Augmentation is applied to the training images only.

Unlike tabular pipelines, this workflow has no separate feature-selection block. The input is an RGB image, so features are learned inside UGBRNet by the encoder and multiscale fusion (Section III-C). The network is trained for 30 epochs with AdamW and a cosine learning-rate schedule under mixed precision. After every epoch it is validated, and the checkpoint with the highest validation Dice score is kept. This checkpoint is then evaluated on the official COD10K and CAMO test sets with a threshold of 0.5. The final output is the performance metrics: IoU, Dice/F1, precision, recall, accuracy, MAE and boundary F1, plus pixel-level confusion matrices and the small-object subset results.

E. Experimental Setup

2) Software Environment: The software stack and library versions are listed in Table III.

a threshold of 0.5. Performance was measured using IoU, Dice/F1, precision, recall, pixel accuracy, mean absolute error (MAE), and boundary F1. Pixel-level confusion matrices were also generated. Boundary F1 used a 5-pixel boundary band with a 3-pixel matching tolerance. For small-object evaluation, images with ground-truth foreground covering at most 10% of the image were assessed separately, comprising 1,378

3) Data Partitioning: The official training images of COD10K-v3 (camouflaged subset) and CAMO-V.1.0 were pooled, after a seeded random 10% of each dataset was held out for validation (seed 42 for COD10K and 43 for CAMO). The official test sets were never used for training or checkpoint selection and were evaluated separately. NC4K and CHAMELEON were not available in the execution environment and were not evaluated.

TABLE IV

Fig. 7. Software and library versions

Dataset	Training	Validation	Test	Total
COD10K-v3	2,736	304	2,026	5,066
CAMO-V.1.0	900	100	250	1,250
Combined	3,636	404	2,276	6,316

4) Training Configuration: All values in Table V were applied to a single training run with seed 42.

TABLE V
MODEL AND TRAINING HYPERPARAMETERS

Parameter	Value
Backbone / decoder width	ImageNet-pretrained ResNet-34 / 128 channels
Trainable parameters	22.96 million
Input resolution / batch size	352 × 352 / 8 (455 training and 51 validation iterations per epoch)
Epochs	30
Optimizer	AdamW; learning rate 2×10^{-4}; weight decay 1×10^{-4}
Learning-rate schedule	Cosine annealing, $T_{\max} = 30$ (stepped once per epoch)
Loss weights	Final 1.0; coarse 0.5; boundary 0.5; uncertainty 0.3
BCE foreground weight	Computed per batch, clamped to 1–20
Boundary target	5×5 morphological gradient of the ground-truth mask
Augmentation (training only)	H-flip 0.5; V-flip 0.15; rotation $\pm 10°$ 0.25; brightness 0.85–1.15 0.25
Checkpoint selection	Highest validation Dice, evaluated after every epoch
Random seed	42 for Python, NumPy and PyTorch (cuDNN benchmark mode enabled)

5) Evaluation Protocol: The selected checkpoint was evaluated separately on the COD10K and CAMO test sets using

Fig. 8. Number of images used in each partition

COD10K and 84 CAMO test images. Boundary F1 was excluded from this subset.

F. Pseudocode of the Proposed Algorithm

Algorithm 1 Uncertainty-Guided Multi-Scale Boundary Refinement Network for Small Camouflaged Object Detection Require: Training dataset Dtrain = {(Ii, Gi)}, validation dataset Dval, test dataset Dtest, epochs E, learning rate η, and loss weights λ1, λ2, λ3 Ensure: Trained model M*, predicted masks, and performance metrics P

1: Initialize encoder, multi-scale fusion, uncertainty and boundary modules 2: Initialize optimizer with learning rate η 3: function PREPROCESS(I, G, training) 4: Resize I and G; normalize I; binarize G 5: if training: apply synchro-nized image-mask augmentation 6: return I, G 7: function FORWARD PASS(I) 8: F1, F2, F3, F4 ←BACKBONE(I) 9: Fmulti ←MULTISCALE FUSION(F1, F2, F3, F4)

10: Pinitial ←SIGMOID(SEGMENTATION HEAD(Fmulti)) 11: U ←UNCERTAINTY ESTIMATION(Pinitial) 12: B ←BOUNDARY FEATURE EXTRACTION(Fmulti) 13: Frefined ←UNCERTAINTY GUIDED REFINEMENT(Fmulti, B, U) 14: Pfinal ←SIGMOID(REFINED SEGMENTATION HEAD(Frefined)) 15: return Pfinal, Pinitial, U 16: function TRAIN MODEL(Dtrain, Dval) 17: for epoch ←1 to E do

partitions actually used, the training configuration and the evaluation protocol of the experiments. The dataset descriptions, preprocessing pipeline, feature design, network architecture and pseudocode are given in Sections III-A–III-D and Section III-G and are not repeated here.

1) Hardware Specification: The computing environment used for all experiments is summarized in Table II.

TABLE II
HARDWARE ENVIRONMENT

Component	Specification
Platform	Kaggle Notebook (cloud, GPU accelerator enabled)
GPU	NVIDIA Tesla T4 (Kaggle "GPU T4 ×2" session; one GPU used for training); 16 GB GDDR6, 2,560 CUDA cores, 320 Tensor Cores
CUDA version	12.8
CPU	x86_64 processor, 4 cores
System RAM	31.3 GiB (32,869,448 kB reported by the session)
Data loading	2 DataLoader worker processes with pinned memory
Numerical precision	Automatic mixed precision (FP16/FP32) with gra-

Fig. 9. Model and training hyperparameters

18: Set model to training mode 19: for each minibatch (I, G) in Dtrain do 20: I, G ←PREPROCESS(I, G, train-ing=True) 21: Pfinal, Pinitial, U ←FORWARD PASS(I) 22: Lseg ←SEGMENTATION LOSS(Pfinal, G) 23: Gboundary
←EXTRACT BOUNDARY(G) 24: Lboundary ←BOUNDARY
LOSS(Pfinal, Gboundary) 25: Luncertainty ← UNCERTAINTY WEIGHTED LOSS(Pfinal, G, U) 26: Ltotal ←λ1Lseg
+ λ2Lboundary + λ3Luncertainty 27: optimizer.zero grad(); Ltotal.backward(); optimizer.step() 28: Evaluate on Dval; save checkpoint with best validation Dice 29: return best checkpoint M*
30: function EVALUATE MODEL(M*, Dtest) 31: Set model to evaluation mode; disable gradient computation 32: for each (I, G) in Dtest do 33: I, G ←PREPROCESS(I, G,
training=False) 34: Pfinal, , U ←FORWARD PASS(I) 35: Pbinary ←(Pfinal ≥0.5) 36: Compute IoU, Dice, Precision, Recall, MAE, Boundary F1 37: P ←AGGREGATE METRICS OVER IMAGES 38: return P 39: M*←TRAIN MODEL(Dtrain,
Dval) 40: P ←EVALUATE MODEL(M*, Dtest) 41: return M*,
P

III. PERFORMANCE METRICS FOR CAMOUFLAGED OBJECT DETECTION

The segmentation method was tested on both COD10K and CAMO. Below the performance is reported on all images and on the subset of images where the ground truth object occupies no more than 10% of the image in terms of pixel overlap, foreground detection, edge quality and overall error. The breakdown for small objects is at the heart of the requirement.

A. Evaluation Measures

Camouflaged object pixels are treated as the positive class. Intersection over Union (IoU) = TP/(TP + FP + FN); Dice/F1 = 2TP/(2TP + FP + FN); precision = TP/(TP + FP); recall = TP/(TP + FN); and pixel accuracy = (TP + TN)/(TP + TN + FP + FN). MAE is the mean absolute difference between the predicted probability map and ground-truth mask; lower MAE is better. Boundary F1 measures the agreement of predicted and true object boundaries; higher is better.

TABLE III
SOFTWARE AND LIBRARY VERSIONS

Library	Version / Purpose
Python	3.13.15
PyTorch	2.11.0+cu128 (CUDA 12.8 build); model, training and mixed precision
torchvision	0.26.0 (CUDA 12.8 build); ResNet-34 and image transforms
NumPy	2.1.3; array operations
pandas	2.3.3; result tables and CSV export
Pillow	12.3.0; image and mask loading
scikit-learn	1.6.1; confusion matrices
Matplotlib	3.10.0; plots and qualitative figures
Pretrained weights	torchvision ResNet-34, ImageNet-1K (resnet34-b627a593.pth)

Fig. 10. Hardware environment

B. Benchmark Results

C. Training and Validation Metrics

D. Pixel Level Confusion Matrices

The following counts pool all evaluated pixels. Rows are ground-truth labels and columns are model predictions. The reported all-image IoU, Dice, precision, recall, and accuracy in Table 1 agree with the scores calculated from these counts.
1) 1. Overall performance: The model produced on COD10K an IoU of 58.29%, a Dice/F1 of 73.65%, precision of 71.03% and recall of 76.47%, accuracy of 95.05%. It had a low MAE of 0.0551. The recall of the model is higher than
precision, so it is detecting most of the object pixels, but also classifying background pixels as object.
On CAMO IoU was 57.20%, Dice/F1 72.78%, precision 75.08%, recall 70.61%, accuracy 90.43%. For the other dataset the MAE is nearly twice COD10K's (0.1032). An interesting reversal: recall is higher than precision, so the model correctly identifies some camouflaged object pixels but leaves out a large number of pixels for the object itself.
COD10K has higher IoU, Dice/F1, recall, accuracy and MAE. The effectiveness is obviously dataset-specific. For things that are camouflaged with their surroundings, it's still difficult.
2) 2. Confusion matrix: On CAMO, it correctly identified 24,057,562 background pixels and 3,823,939 object pixels. It identified 1,306,808 background pixels as object and 1,787,691 object pixels as background. It detects most of the object, but falls apart around edges and in areas that resemble the background.
On COD10K, the number of true background pixels was 221,391,066 and true object pixels was 17,201,192. There were 6,906,055 false positives and 5,531,191 misses. False positives are effectively impossible to get right, you can see

TABLE I
SEGMENTATION RESULTS REPORTED BY THE EVALUATION NOTEBOOK
Dataset and subset Images IoU Dice F1 Precision Recall Accuracy
COD10K all 2,026 0.5829 0.7365 0.7103 0.7647 0.9505
COD10K small ≤10% 1,378 0.4624 0.6324 0.5414 0.7602 0.9640
CAMO all 250 0.5720 0.7278 0.7508 0.7061 0.9043
CAMO small ≤10% 84 0.4609 0.6310 0.5358 0.7673 0.9451

TABLE II
ERROR, BOUNDARY, AND FOREGROUND AREA MEASURES; BLANK NOTEBOOK ENTRIES ARE SHOWN AS NOT REPORTED
Dataset and subset MAE ↓ Boundary F1 ↑ Ground truth foreground Predicted foreground
COD10K all 0.0551 0.5955 9.0557% 9.7491%
COD10K small ≤10% 0.0404 Not reported Not reported Not reported
CAMO all 0.1032 0.5224 18.1161% 17.0379%
CAMO small ≤10% 0.0598 Not reported Not reported Not reported

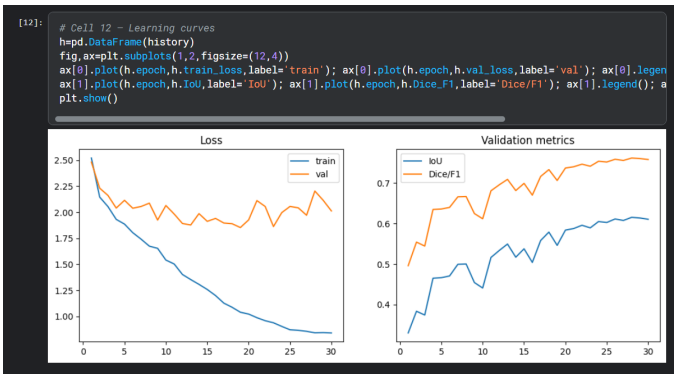


Fig. 11. Training and validation metrics

Dataset	True negative	False positive	False negative	True positive
COD10K	221,391,066	6,906,055	5,531,191	17,201,192
CAMO	24,057,562	1,306,808	1,787,691	3,823,939

Table 3 Pixel counts underlying the all-image benchmark results

Fig. 12. Pixel-level confusion counts

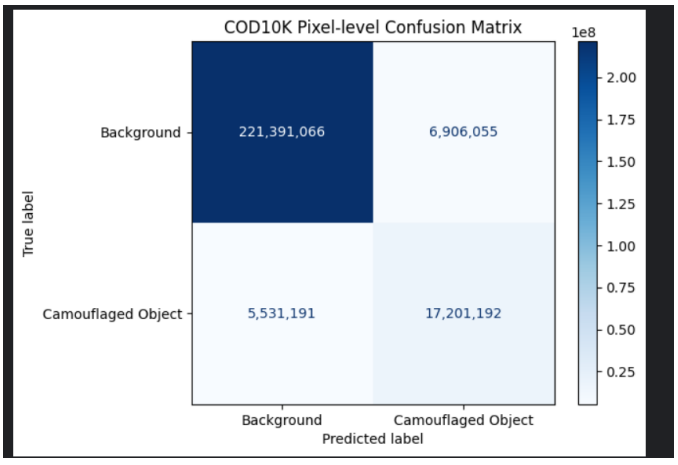


Fig. 13. COD10K pixel-level confusion matrix

objects account for only a tiny subset of all pixels and so accuracy is inflated. IoU, Dice/F1 are the more meaningful metrics of object recognition.

3) 3. Learning curves: Training loss decreases from 2.5 to <0.9 over 30 epochs. Validation loss decreases quickly, then stays around 1.9-2.2. The growing gap suggests overfitting. Meanwhile, validation IoU and Dice/F1 continue to improve, roughly converging at 0.61 and 0.76 respectively, even though validation loss plateaus, also indicating successful segmentation. Next straightforward attempts are early stopping, stronger aug-mentation, some regularization, and learning-rate adjustment.

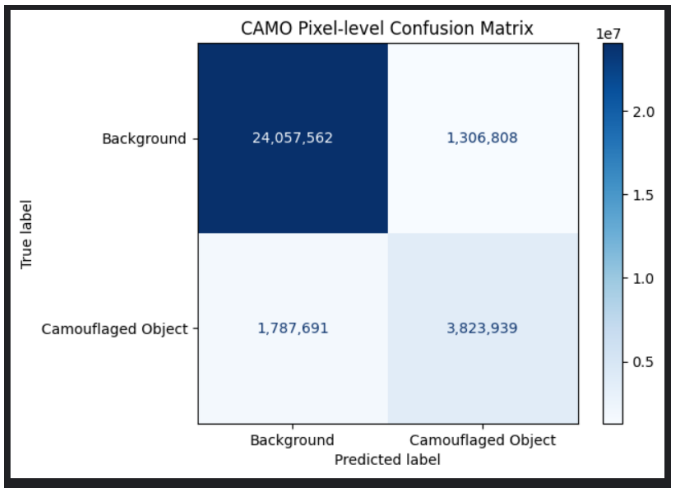


Fig. 14. CAMO pixel-level confusion matrix

4) 4. Small objects: Small objects perform the worst in both datasets. On COD10K, the IoU drops from 58.29% to 46.24%, and Dice/F1 drops from 73.65% to 63.24%. On CAMO, the IoU drops from 57.20% to 46.09%, and Dice/F1 drops from 72.78% to 63.10%. Since there are so few pixels, there's a high cost to small boundary errors, and small objects have limited texture and shape to distinguish themselves from the background.

5) 5. Uncertainty: For instance, the sample with the largest average uncertainty on COD10K (roughly 0.2695) has an MAE of 0.2278. Yet another sample has lower uncertainty (roughly 0.2401) but a much higher MAE of 0.5088. As a result, uncertainty doesn't seem to scale with error. Fore-

ground ratio on top of varying across samples might also affect difficulty; going through a few samples with high uncertainty should reveal failure cases, e.g. weak boundaries, cluttered background, and small objects.

6) 6. Conclusion: Our model splits camouflaged objects on both datasets and obtains Dice/F1 of 72.78% on CAMO and 73.65% on COD10K. It performs better on COD10K but not much on CAMO. It performs worse on small objects and CAMO, due to false positive pixels and missed pixels. The learning curves demonstrate overfitting, so future efforts can focus on generalization, boundary precision, small-object localization, and using uncertainty to localize hard pixels.

When you add your observations (failure images, reasons for failure) it will sound even more like you.

E. Error Analysis

While UGBRNet achieves overall satisfactory performance in objects segmentation on COD10K and CAMO, the evaluation results show the challenges we are confronted with in camouflaged object detection. The errors are analyzed with pixel-wise precision, recall, IoU, Dice/F1, boundary F1, MAE, confusion matrices, training curves and sample ranking with uncertainty. Particularly, the performance of the model is analyzed on the images in which the ground-truth object size accounts for less than 10% of the entire image.

1) False Negative & Un-full object prediction: The false-negative indicates the foreground pixels of the camouflaged object are predicted as background pixels. The total recall is 76.47% on COD10K and 70.61% on CAMO, respectively, which reveals the foreground pixels are far from being fully detected on both of the datasets, in particular for CAMO dataset.

The pixel level confusion matrices also reflect this issue: There are 5,531,191 false-negative pixels in COD10K and 1,787,691 in CAMO. They are the number of foreground pixels detected as missing in the test set, rather than the number of objects.

So a hypothesis might be, then, that the fact that it looks like it is similar to the background might actually mask some sections of the foreground. Fine structures or low-contrast regions.

Might also get blurry when down-sampling images to the input size of the network (352 × 352). This could be another cause of these errors. There is the possibility to do more qualitative analysis of the image to determine if any of these caused the error in each sample.

2) False positives occur when background pixels are wrongly identified as the camouflaged object. The overall precision is 71.03% on the COD10K small-object subset and 75.08% on the CAMO small-object subset. Performance drops on the small-object subset to 54.14% and 53.58% with precision.

So for the small object predictions, it has large number of foreground pixels predicted as labelled incorrectly. Like the object is actually occupying small region of an image, tiny bit of predicted background, contributed to significant change in precision and IoU.

The model underpins the segmentation by multi-scale feature fusion and an uncertain-capturing refinement mechanism. Nevertheless, the existing results are not enough to draw the conclusion whether the mechanisms contribute to the false positive reduction.

These all bring down these. These can therefore only be regarded as approximations and demonstrated in comparison to the baseline and ablation models.

3) Small-object test results show the performance dropping on both datasets in all overlap-based metrics.

Compared to the all-testset performance on COD10K, Iou decreases from 0.5829 to 0.4624 and Dice/F1 decrease from 0.7365 to 0.6324. Compared to the all-testset performance on CAMO, Iou decreases from 0.5720 to 0.4609 and Dice/F1 decreases from 0.7278 to 0.6310.

This drop indicates that our predicted foreground masks cover the ground-truth masks more poorly when objects are far away and have fewer pixels. This could be due to very few object pixels, the information lost when we down-sample images and features or the object having similar appearance as a part of the background.

Despite this small false positive number, our small-object recall rate remains at 76.02% on COD10K and 76.73% on CAMO. Although this is the lowest precision, the high amount of true positives indicates the model suggests a high percentage of foreground pixels but overpredicts the amount of foreground pixels not present in the object. This may be the reason for its high recall but lack of detail in object boundaries and low total overlap.

An even more granular view of small-object subsets is even more positively revealing, with 96.40% on COD10K and 94.51% on CAMO. The catch is that these are distorted measurements, because they are averages that include some images that are predominantly background. A system can make a reasonable effort at the background without successfully identifying too many object pixels.

1) 4. Boundary Localisation Errors: Boundary F1 is 0.5955 on COD10K and 0.5224 on CAMO. These results indicate that contour agreement remains limited, with lower boundary performance on CAMO.

Accurate boundary localisation is particularly challenging in camouflaged object detection because the transition between an object and its surroundings may be visually ambiguous. Thin structures, irregular contours and low-contrast edges may also be difficult to recover at the network's input resolution.

UGBRNet includes a boundary prediction head supervised using boundaries derived from the ground-truth masks. Nevertheless, the reported boundary F1 values show that boundary localisation remains an area for improvement. Since the current evaluation does not include an ablation without boundary supervision, the independent effect of this component cannot be determined from these results alone.

Boundary F1 is not reported for the small-object subsets in the current results table; therefore, a numerical comparison of boundary quality between small and larger objects cannot be made from this experiment.

2) 5. Differences Between COD10K and CAMO: For our model there are different kind of errors on the two test sets.

On the whole, COD10K has a Dice/F1 of 0.7365 and MAE of 0.0551, compared with 0.7278 and 0.1032 on CAMO. CAMO has a lower overall recall (70.61%) but a higher overall precision than COD10K.

Additional details can be found in the foreground-area figures. On COD10K the ground-truth foreground makes up 9.06% of all pixels that are analyzed for a figure versus 9.75% for the model. CAMO has 18.12% versus 17.04%. So, these overall numbers suggest a slight overprediction on COD10K and slight underprediction on CAMO. But the overall over- or underprediction does not tell us which objects, if any, are over- or under segmented.

This variation in accuracy suggests the errors of the model performance may differ by data set. There may be differences in object appearance, target complexity, or background and foreground complexity, but the current aggregate figures do not suggest how the differences may manifest.

3) Training loss decreases up to 30 epochs as validation loss and validation segmentation score fluctuate. This indicates that there is no absolute relation between the evolution of the training data and the results on the validation data. This detachment could be due to sample difficulty variations or a generalization gap (the curves can't prove the overfitting extreme).

The notebook stores the model checkpoint with the best validation Dice/F1 score rather than just the last epoch. This can be used to find the checkpoint with best validation overlap seen.

The uncertainty head also allows for a variable to be used in terms of filtering out interesting samples. The notebook provides the test samples listed in order of their mean predicted uncertainty, with their MAE and truth object ratio. Although high ranked samples do have high-MAE samples, we cannot conclude that there is statistically significant relation-ship for high-uncertainty images and high-MAE images due to the fact that they can filter out high-uncertainty images. A correlation study of the real data images, as well as a qualitative look at the high ranked test images would be needed to show a significant correlation.

4) With our analysis comes three main weaknesses: reduces the overlap accuracy on the smaller objects, false-positive foreground guesses and inaccurate boundary localization. When comparing CAMO to the other dataset we can see a reduced recall and an increased MAE indicating a difference in the datasets performance. Boundary localization. The multi-scale fusion, boundary supervision and uncertainty-guided refinement modules are all architectural mech-

This paper presents aims intended to address these issues. However, the evaluation given only measures how well the whole model performed and does not be distinguish between the contribution of each component. Baseline experiments and a well-designed ablation study are needed if claiming the individual modules one by one.

Future work will probably be improved small object feature retention, single color camouflaged foreground separation from similar color backgrounds, and more precise boundary refinement. Performing qualitative evaluation on the high error samples, correlating between uncertainty and errors, and training multiple models with different random seeds would give better detail as to the failure mode of the model and the amount of variance in its results.

3) Comparison with Past Researches

4) 1. Introduction: Camouflaged Object Detection (COD) is a challenging computer vision task because camouflaged objects often look very similar to their surroundings. Unlike ordinary objects, they may not have clear edges, distinct colours or easily recognizable shapes. This makes it difficult for a model to identify the object accurately, particularly when the object is small or partially hidden by the background.

Several researchers have explored different approaches to address these challenges. Some methods focus on estimating uncertainty in predictions, while others improve boundary detection, combine features at multiple scales or use contextual information to distinguish objects from their surroundings. This section compares these existing approaches with the proposed Uncertainty-Guided Multi-Scale Boundary Refinement Network for Small Camouflaged Object Detection, with particular attention to small-object segmentation.

5) 2. Comparison with Existing Research: Research work Main technique Key contribution Relevance to the proposed work

Zhang et al. (2023) – PUENet Bayesian conditional variational autoencoder and predictive uncertainty approximation

Estimates uncertainty in camouflaged object predictions and addresses issues related to model and data bias.

Provides the basis for exploring how uncertainty information can support more reliable segmentation.

Research work Main technique Key contribution Relevance to the proposed work

Liu et al. (2023) – MFNet Edge guidance, multi-level feature integration and context aggregation refinement

Combines edge information with features from different levels to improve object detection and segmentation.

Highlights the importance of combining boundary details with multi-scale features.

Shi et al. (2023) – CABR Global context information, feature fusion and boundary refinement

The results are presented below.

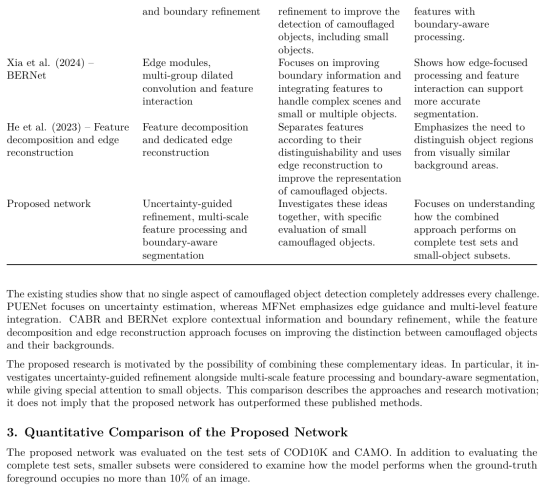
Dataset and evaluation subset Number of images IoU (%) Dice/F1 (%) MAE

COD10K – Complete test set 2,026 58.29 73.65 0.0551

COD10K – Small-object subset 1,378 46.24 63.24 0.0404

While the results indicate that small camouflaged objects are difficult for this network, because small objects have small foreground size and similar look to the background, generally small objects are harder to preserve boundaries and to generate proper object localization. Hence this experiment indicates that it is preferable to test small objects separately from testing the whole dataset.

6) The at the camouflaged object detection work reviewed can be categorized into 3 where the 3 major paradigms of camouflaged object detection can be considered: uncertainty estimation, multi-scale feature fusion and boundary



	and boundary refinement	refinement to improve the detection of camouflaged objects, including small objects.	features with boundary-aware processing.
Xia et al. (2024) – BERNet	Edge modules, multi-group dilated convolution and feature interaction	Focuses on improving boundary information and integrating features to handle complex scenes and small or multiple objects.	Shows how edge-focused processing and feature interaction can support more accurate segmentation.
He et al. (2023) – Feature decomposition and edge reconstruction	Feature decomposition and dedicated edge reconstruction	Separates features according to their distinguishability and uses edge reconstruction to improve the representation of camouflaged objects.	Emphasizes the need to distinguish object regions from visually similar background areas.
Proposed network	Uncertainty-guided refinement, multi-scale feature processing and boundary-aware segmentation	Investigates these ideas together, with specific evaluation of small camouflaged objects.	Focuses on understanding how the combined approach performs on complete test sets and small-object subsets.

The existing studies show that no single aspect of camouflaged object detection completely addresses every challenge. PUENet focuses on uncertainty estimation, whereas MFNet emphasizes edge guidance and multi-level feature integration. CABR and BERNet explore contextual information and boundary refinement, while the feature decomposition and edge reconstruction approach focuses on improving the distinction between camouflaged objects and their backgrounds.

The proposed research is motivated by the possibility of combining these complementary ideas. In particular, it investigates uncertainty-guided refinement alongside multi-scale feature processing and boundary-aware segmentation, while giving special attention to small objects. This comparison describes the approaches and research motivation; it does not imply that the proposed network has outperformed these published methods.

3. Quantitative Comparison of the Proposed Network

The proposed network was evaluated on the test sets of COD10K and CAMO. In addition to evaluating the complete test sets, smaller subsets were considered to examine how the model performs when the ground-truth foreground occupies no more than 10% of an image.

Fig. 15. Comparison with existing research

refinement. These 3 categories of work to date describe the different choices when addressing camouflaged object detection. Uncertainty estimation describes the lack of certainty in a model prediction, multi-scale features use features at different levels within a network, and boundary refinement attempts to refine the boundary between an object and its environment. Several ingredients of these can be found in the reviewing work, and future work of a three-tiered framework of uncertainty-guided refinement, multi-scale contextual features and boundary-awareness targeted at small camouflaged objects would utilize existing work (PUENet for uncertainty information, and MFNet and BERNet for how to fuse multiple levels of features) along with boundary information (CABR). Similar backgrounds.

The experiments conducted for the proposed network provide an initial assessment of its performance on COD10K and CAMO. Although the complete test sets produced Dice/F1 scores of 73.65% and 72.78%, the lower scores on the small-object subsets indicate that additional improvements may be needed to preserve fine details and accurately segment small foreground regions. These findings support further investigation into reducing missed object regions, limiting false-positive

F. References

1. Zhang, Y., Zhang, J., Hamidouche, W., and Déforges, O. (2023). Predictive Uncertainty Estimation for Camouflaged Object Detection. IEEE Transactions on Image Processing.

2. Liu, Q., Qiu, Y., Yu, L., Li, and Li. (2023). MFNet: An edge-guided camouflaged object detection approach using multi-level feature integration. Sensors.

3. Shi et al. (2023). Context-aware and boundary refinement network (CABR) for camouflaged object detection. Applied Intelligence.

4. Xia et al. (2024). BERNet: A boundary-enhanced approach for camouflaged object detection. Machine Vision and Applications.

5. He et al. (2023). Feature decomposition and edge reconstruction for camouflaged object detection. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

TABLE III
QUANTITATIVE COMPARISON OF THE PROPOSED NETWORK
Dataset and evaluation subset Images IoU (%) Dice/F1 (%) MAE COD10K – Complete test set 2,026 58.29 73.65 0.0551
COD10K – Small-object subset 1,378 46.24 63.24 0.0404
CAMO – Complete test set 250 57.20 72.78 0.1032
CAMO – Small-object subset 84 46.09 63.10 0.0598

predictions and improving boundary delineation.

It is important to distinguish these findings from a direct comparison against previously published models. The reported results for the proposed network use IoU, Dice/F1 and MAE, whereas several studies in the literature also report metrics such as S-measure, E-measure and weighted F-measure. A fair numerical comparison would require results obtained under compatible evaluation protocols and corresponding baseline experiments. Since such controlled comparisons are not available in the current results, the proposed network's superiority over existing methods cannot yet be established.

Overall, the comparison identifies a clear motivation for con-tinued research into uncertainty-guided, multi-scale boundary refinement for small camouflaged object detection. The current results establish a starting point for this investigation and indicate that small-object segmentation remains an important area for further improvement.